PROJECT TEAM

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ONLINE HEALTHCARE MANAGEMENT SYSTEM

Care that stays connected.

A role-based web portal for appointment coordination, clinical notes and patient care.

ONE WORKSPACE, THREE ROLES

Make everyday care easier to coordinate.

Patients, clinicians and administrators each get tools designed around the work they need to do.

01

Patient

Manage personal care without losing track of the next step.

Book visits | View history | Update profile | Leave feedback

02

Doctor

Coordinate availability and keep visit documentation connected to appointments.

Publish open times | Manage visits | Write notes | Review ratings

03

Administrator

Set up the clinic workspace and oversee scheduling activity.

Manage accounts | Add doctors | Clinic settings | Activity overview

Design goal Keep the common healthcare tasks visible, direct and scoped to the signed-in role.

FROM OPEN TIME TO VISIT HISTORY

A connected appointment journey.

Availability, slot validation and visit documentation keep scheduling tied to care delivery.

01

Publish

Doctor adds a future availability window.

02

Request

Patient chooses a clinician, time and reason.

03

Validate

System checks the window and overlapping bookings.

04

Coordinate

Doctor confirms, completes or cancels the visit.

05

Document

Doctor records a note; patient can review and rate the visit.

Data follows the visit Medical notes are linked to the appointment and patient, with access limited by role.

JAVA WEB ARCHITECTURE

Simple layers, clear responsibilities.

Servlets handle requests, repositories own JDBC operations, and JSP renders role-specific pages.

PRESENTATION

JSP & CSS

Responsive pages, forms and role-specific navigation. Views stay under `WEB-INF`.

↓

WEB & DOMAIN

Jakarta Servlets

Authentication, validation, authorization, workflow rules and request routing.

↓

DATA ACCESS

JDBC repositories

Prepared SQL statements, role-scoped queries and MySQL persistence.

Java
JDK 17+ target

Maven
Build & dependencies

Jetty 12
Local Servlet container

JDBC
MySQL Connector/J

MySQL 8
Relational database

Separation of concerns Models are small Java records; repositories encapsulate database operations; JSP focuses on presentation.

MYSQL RELATIONAL MODEL

Records stay connected to their owners.

Foreign keys keep appointments, clinical notes and feedback attached to the correct accounts.

users — ACCOUNT

id	PK
full_name	required
email	unique
password_hash	PBKDF2
role, specialty, active	access

appointments — VISIT

id	PK
patient_id	FK: users
doctor_id	FK: users
appointment_at	datetime
status, reason	workflow

doctor_availability — SCHEDULE

id	PK
doctor_id	FK: users
available_from	datetime
available_to	datetime
published window	slot rule

users 1 : many appointments 1 : many medical_records 1 : many feedback

medical_records — CLINICAL NOTE

id	PK
appointment_id	FK
patient_id, doctor_id	FK: users
diagnosis	required
notes	private

feedback — REVIEW

id	PK
patient_id, doctor_id	FK: users
rating	1 to 5
comments	optional
created_at	timestamp

system_settings — CONFIG

setting_key	PK
setting_value	clinic options
clinic name	configurable
appointment length	10-180 min

RUBRIC-ALIGNED IMPLEMENTATION

Built to demonstrate the core Java web skills.

1 **Problem understanding**

Three roles have distinct dashboards and permitted actions.

2 **Core Java concepts**

Models, services, collections, validation and exception handling keep responsibilities explicit.

3 **JDBC integration**

Prepared statements and repository classes connect Java operations to MySQL tables.

4 **Servlets & web integration**

Servlet routes, session authentication, JSP views and a responsive interface.

5 **Safer account handling**

Salted PBKDF2 hashes, session ID rotation, CSRF validation and role checks.

6 **Readable delivery**

Modular Maven layout, schema script, setup README and focused unit tests.

Run locally

- JDK 17+, Maven 3.9+, MySQL 8+
- Load `database/schema.sql`
- Set DB environment variables
- `mvn clean test package`

Submission checks

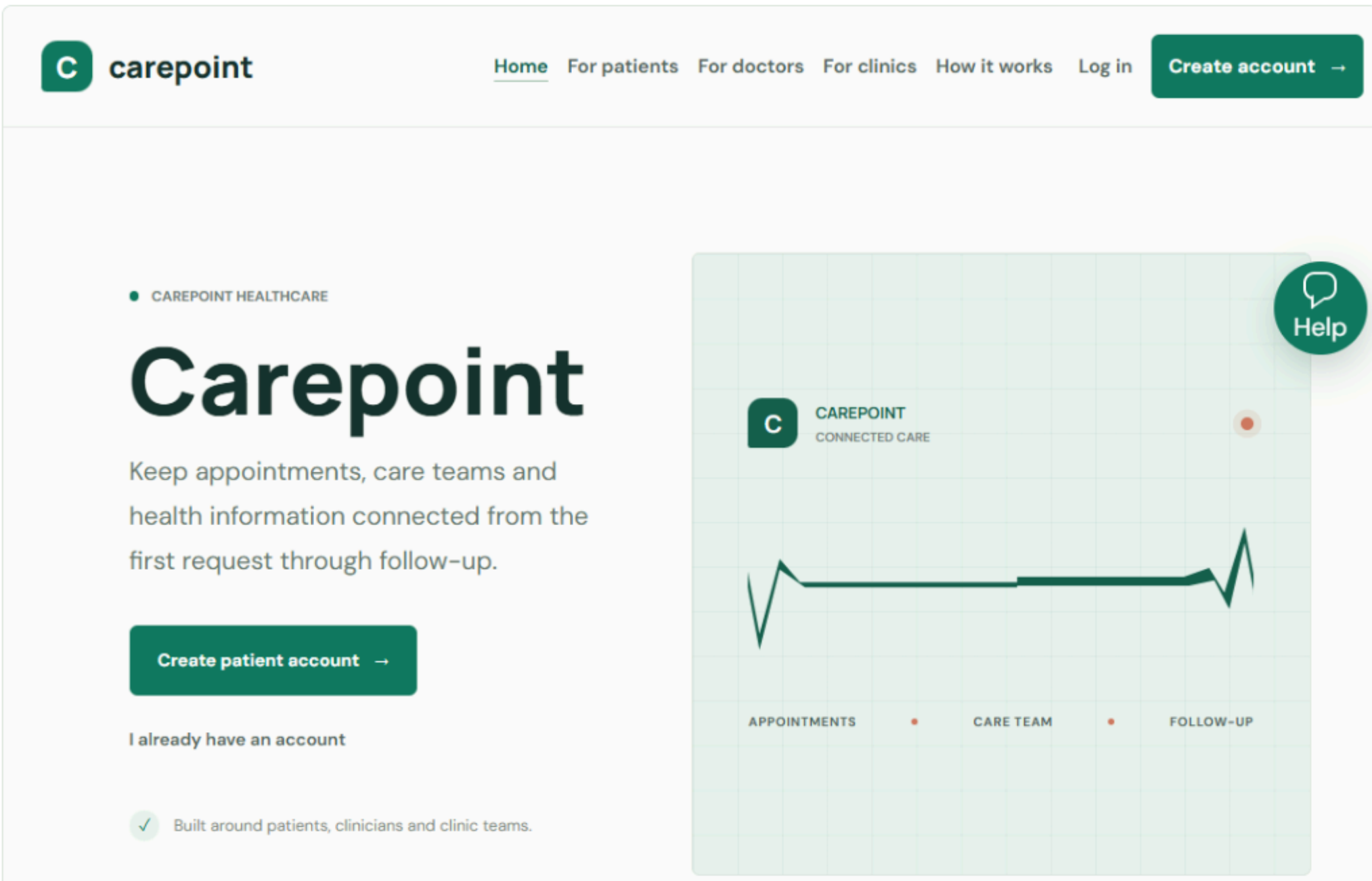
- Push current source to a reviewer-accessible GitHub repository
- Keep this README with setup and run steps
- Include this PDF deck and relevant app screenshots
- Verify repository link and deadline requirements

Carepoint Healthcare Academic demonstration only. Not intended for real patient data or clinical use.

PUBLIC PORTAL

A clear first impression.

The landing page explains the value of the system and provides direct access to account creation and sign-in.



ONE PLACE FOR EVERY ROLE

The details of care, in sync.

Choose your Carepoint experience to see how it supports your day.

01 PATIENTS

Find and follow your care

Appointments, shared records and your profile.

→

02 DOCTORS

Coordinate your practice

Availability, patient visits and clinical notes.

→

03 CLINICS

Manage clinic operations

Accounts, settings and appointment activity.

→

See how Carepoint works →

FROM REQUEST TO FOLLOW-UP

One clear care journey.

Each step moves to the person responsible, keeping appointments and visit information together.

01

Choose an open time

Doctors publish availability. Patients request a visit with a preferred time and reason.

02

Coordinate the appointment

Doctors review requests and confirm, reschedule or update the appointment.

03

Keep the follow-up connected

Doctors add visit notes; patients can review shared records and maintain personal history.

Explore the complete workflow →

carepoint

Care, thoughtfully coordinated.

Sign in →

What visitors see first

Clear messaging, patient and clinician roles, a responsive feature overview, and a prominent account creation path.

Responsive design

Core public content adapts to desktop and mobile views.

Clear navigation

How it works, features, security and support links stay accessible.

Accessible actions

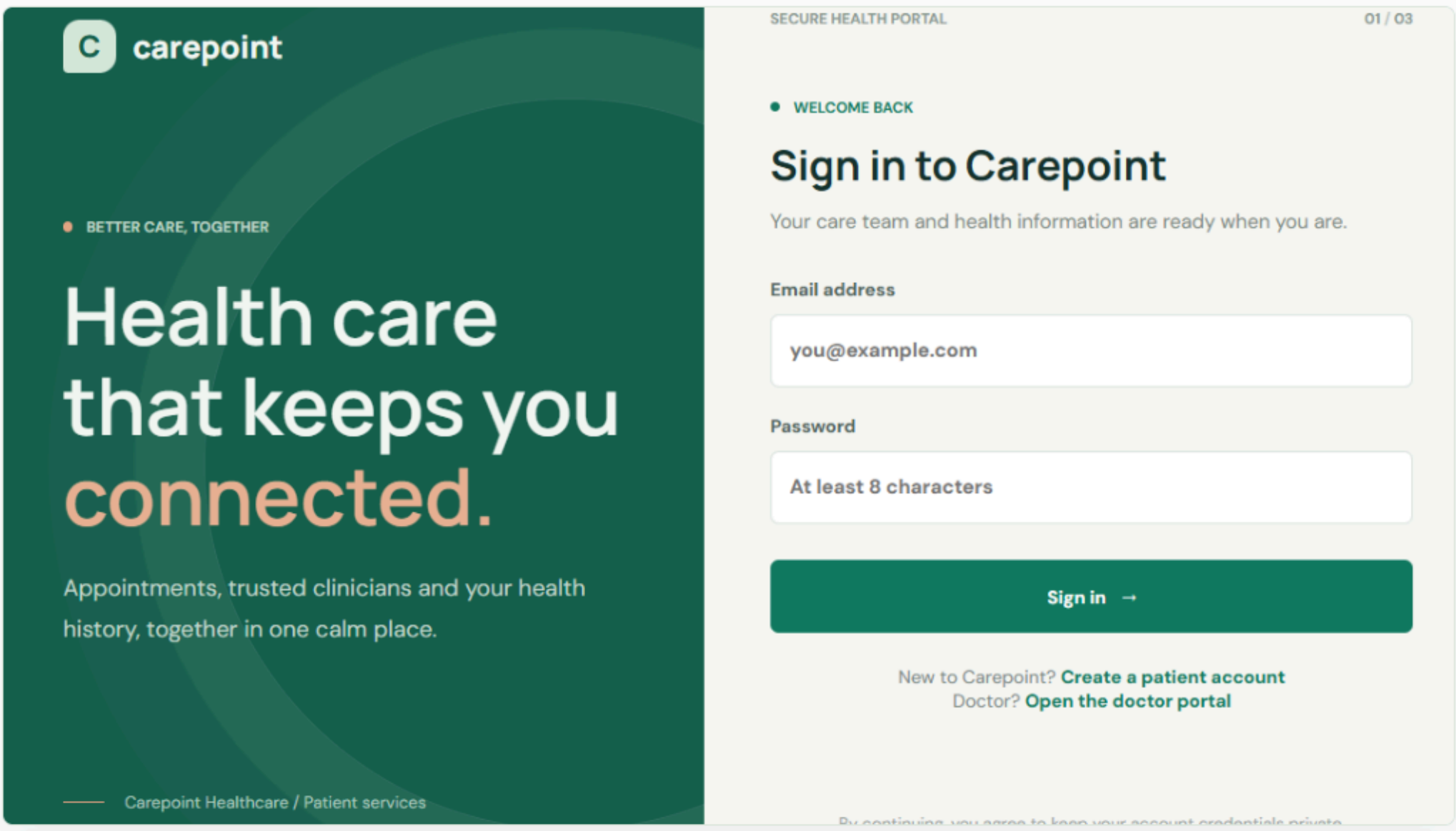
Primary sign-in and registration actions use consistent labels and structure.

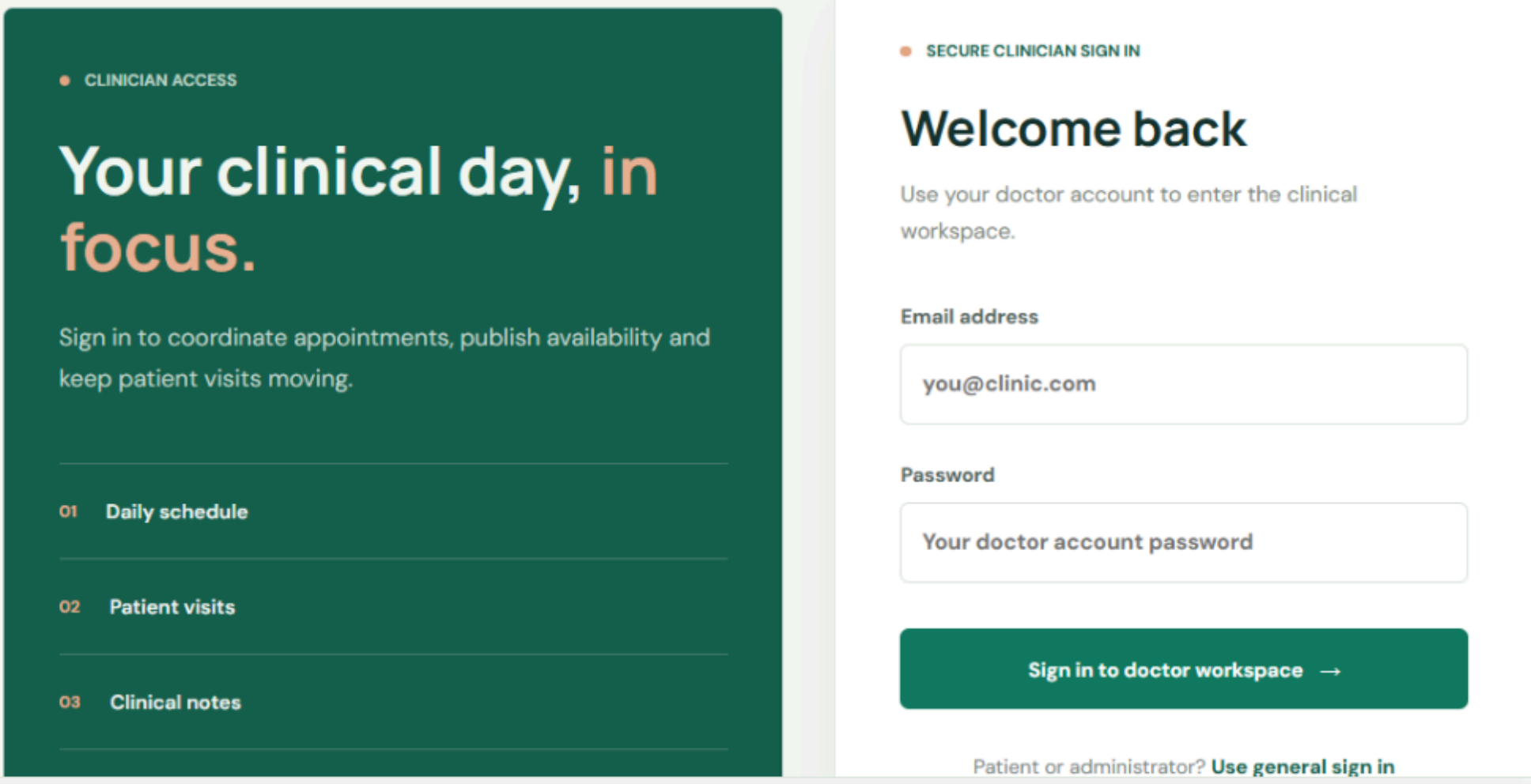
SCREENSHOT: docs/screenshots/landing-page.png

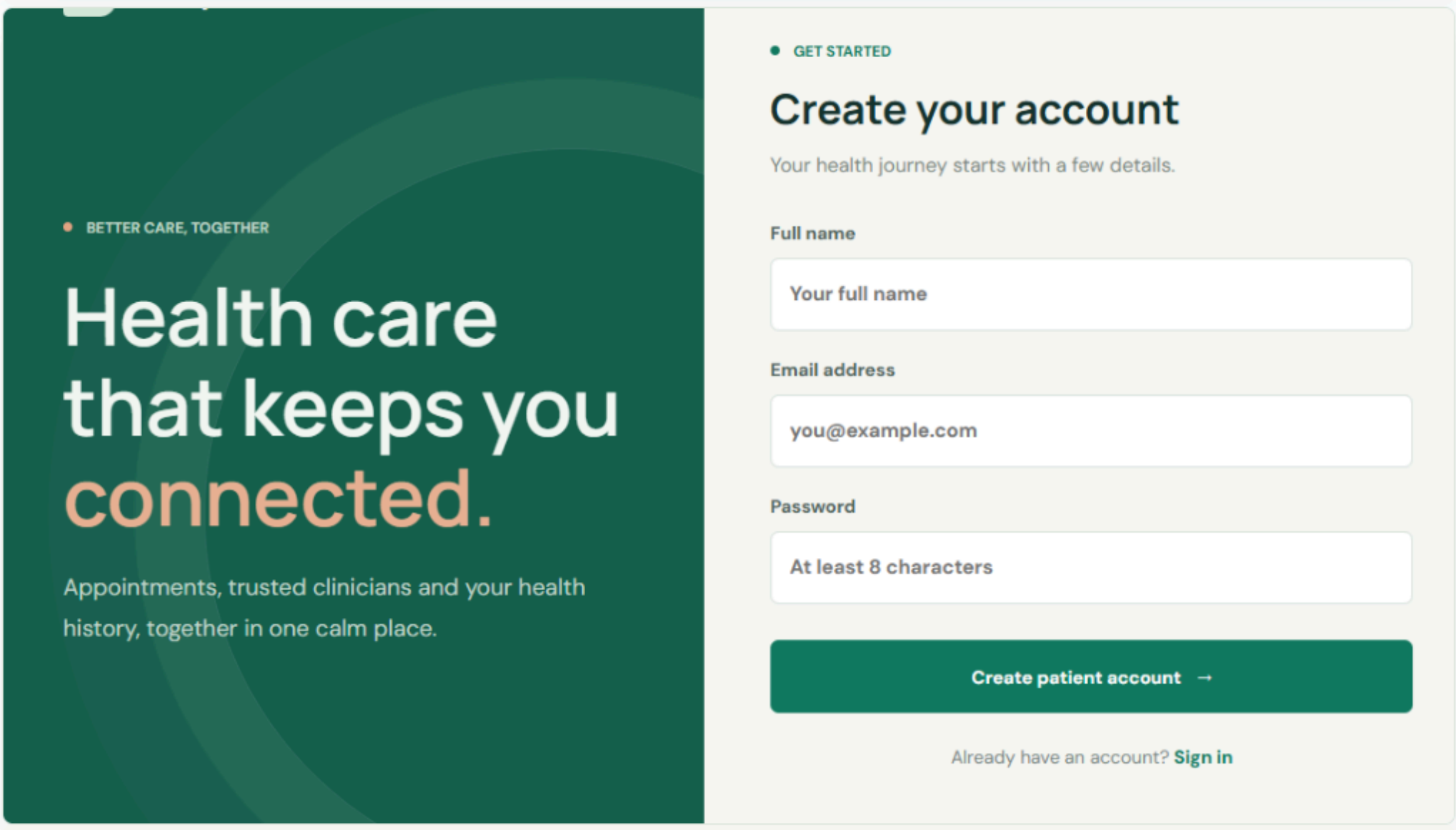
AUTHENTICATION

Role-aware sign-in.

Each role receives a focused login experience, while the shared layer preserves secure account handling.







Security baseline

Password hashes use salted PBKDF2, sessions rotate after authentication, and CSRF validation is applied to state-changing requests.

PATIENT WORKSPACE

Care at a glance.

Patients can review upcoming appointments, access shared records, and move into schedule or profile actions.

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Home For patients For doctors For clinics How it works Log in Create account →

FOR PATIENTS

Your care, easier to follow.

Plan appointments, keep track of visits and find the medical history your care team has shared.

Help

Create a patient account →

Already registered? Sign in

YOUR PATIENT WORKSPACE

Keep every next step close.

01 **Request an appointment**

Choose a doctorâ□□s published time, add a reason and send a request.

02 **Follow or change your visit**

Check appointment status, cancel, or choose another available time for a pending or confirmed visit.

03 **Keep your history together**

Read notes shared by doctors and add personal entries for details you want to remember.

04 **Manage your account**

Update contact details, change your password and leave feedback after a completed visit.

YOUR SIDE OF THE JOURNEY

Know what happens next.

01 / REQUEST

Ask for a visit

Pick an open doctor time and briefly describe the reason for your appointment.

02 / CONFIRM

Track the status

Your request appears in appointment history while the doctor reviews and updates it.

03 / FOLLOW UP

Keep your information current

Shared clinical notes and your own health-history entries are available from Records.

carepoint

Care, thoughtfully coordinated.

Sign in →

Patient focus

Appointment status, medical record access and available clinician discovery are grouped into clear, role-specific views.

SCREENSHOT: docs/screenshots/patient-page.png

CLINICAL WORKSPACE

Every visit, organized.

Doctors can publish availability, review scheduled visits, record clinical notes and manage patient-facing follow-up.

FOR DOCTORS

Your practice, in view.

Coordinate appointment availability, manage patient visits and keep clinical documentation connected.

Help

Sign in to your doctor workspace →

YOUR DOCTOR WORKSPACE

A clear view of your day.

01 **Publish availability**

Choose a date and time window; patients can request visits only inside open times.

02 **Coordinate appointments**

Use the upcoming agenda to review requests, confirm visits and move eligible appointments to another open time.

03 **Document and update visits**

Add diagnoses and notes linked to an appointment, then edit your own note if it needs correction.

04 **Review patient feedback**

See ratings and comments shared after completed appointments.

CLINICAL WORKFLOW

From open slot to documented visit.

Keep the schedule and the patient record connected throughout the visit.

01 / PLAN

Set open times

Publish future windows that match the time you can see patients.

02 / REVIEW

Manage your agenda

See pending and confirmed appointments, and keep your schedule current.

03 / DOCUMENT

Close the loop

Record visit notes and see patient feedback after the visit is completed.

Clinical focus

Visit workflows and record actions are separated from patient account management and platform administration.

SCREENSHOT: docs/screenshots/doctor-page.png

PLATFORM OPERATIONS

Administration without the noise.

Administrators can manage users, clinics, appointment activity, platform settings and operational reporting.

carepoint Home For patients For doctors For clinics How it works Log in Create account →

FOR CLINICS

A clearer view of clinic operations.

Give administrators one place to coordinate people, appointment activity and clinic settings.

Help

Sign in to your clinic workspace →

YOUR ADMIN WORKSPACE

Keep the clinic moving.

01 **Manage people and roles**

Create doctor accounts, update eligible roles and contact details, and control access.

02 **Oversee appointments**

Review clinic-wide visits, update status and reschedule pending or confirmed appointments into open time.

03 **Configure the clinic**

Set the clinic name and the default appointment duration used for new patient requests.

04 **Review activity trends**

Compare six months of appointment volume and account registrations with current appointment statuses.

ADMINISTRATOR VIEW

One workspace for clinic operations.

Operational actions are role-protected; accounts with clinical history are retained rather than hard-deleted.

ACCESS

People and roles

Review accounts, update details and deactivate access when needed.

SCHEDULING

Appointment oversight

Monitor visit status across the clinic and reschedule eligible bookings.

ACTIVITY

Monthly trends

Review recent appointment and account-registration counts over time.

carepoint Care, thoughtfully coordinated. Sign in →

Platform focus

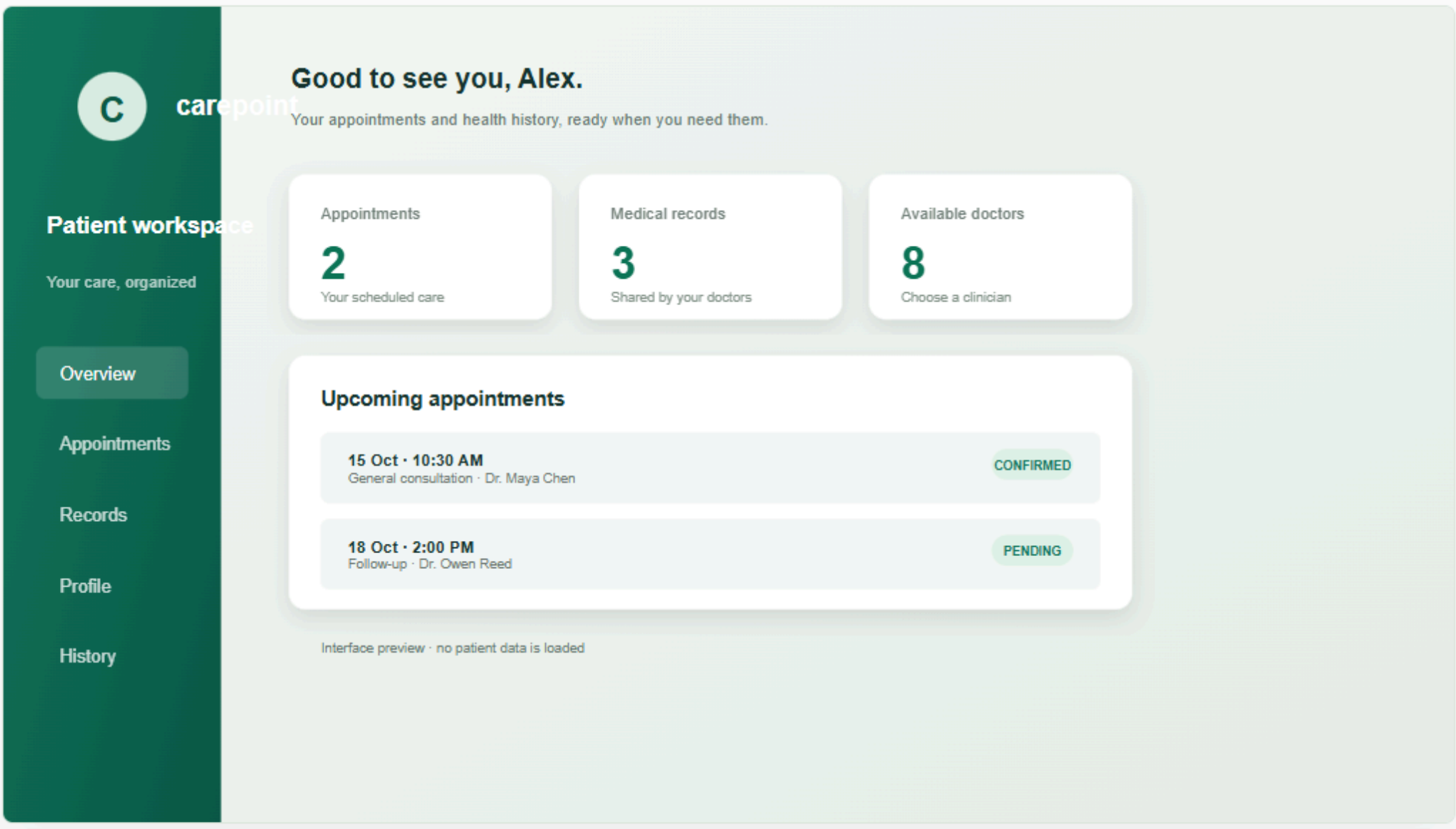
Clinic configuration and user management sit alongside the system-wide activity overview.

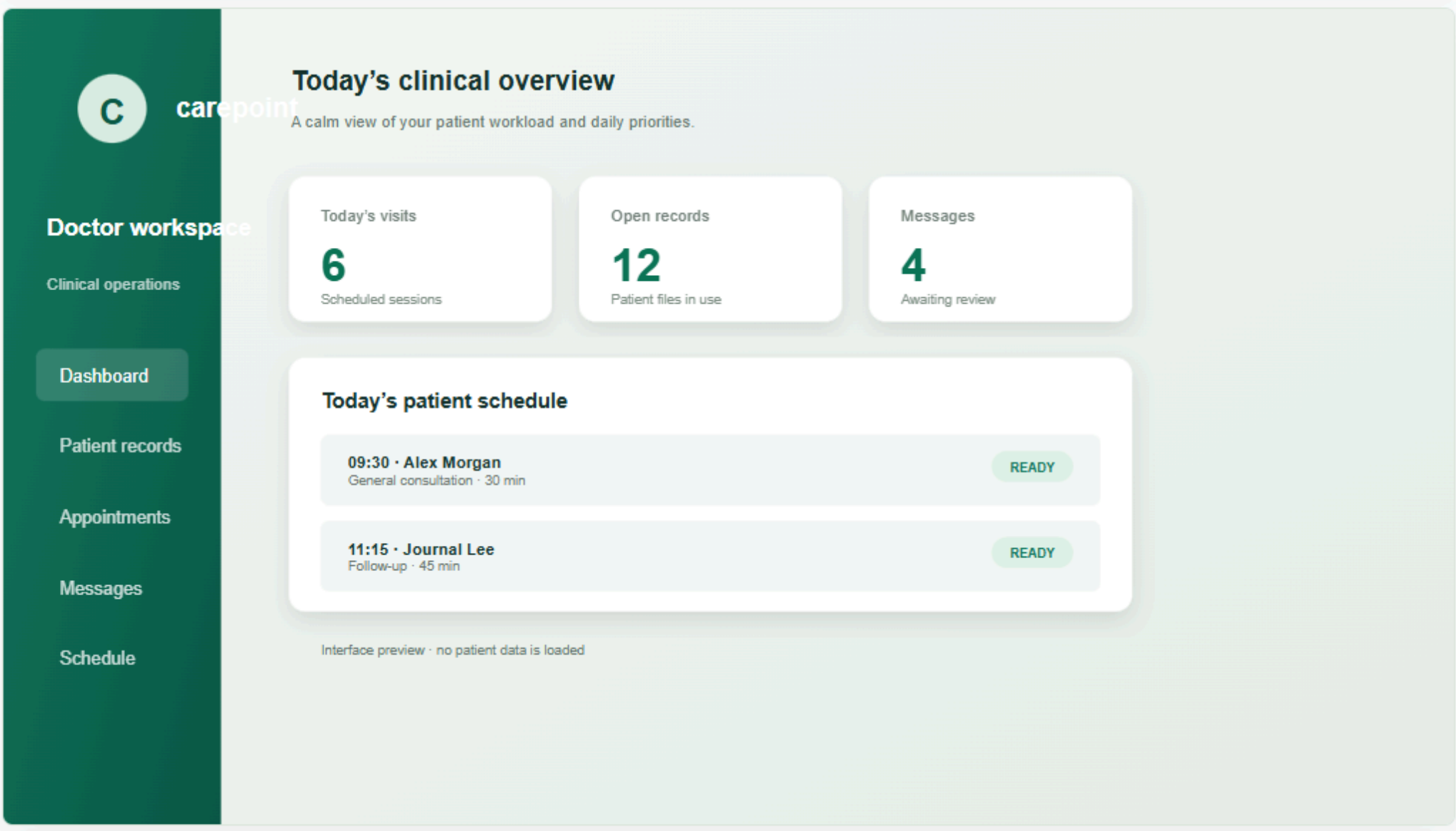
SCREENSHOT: docs/screenshots/clinic-page.png

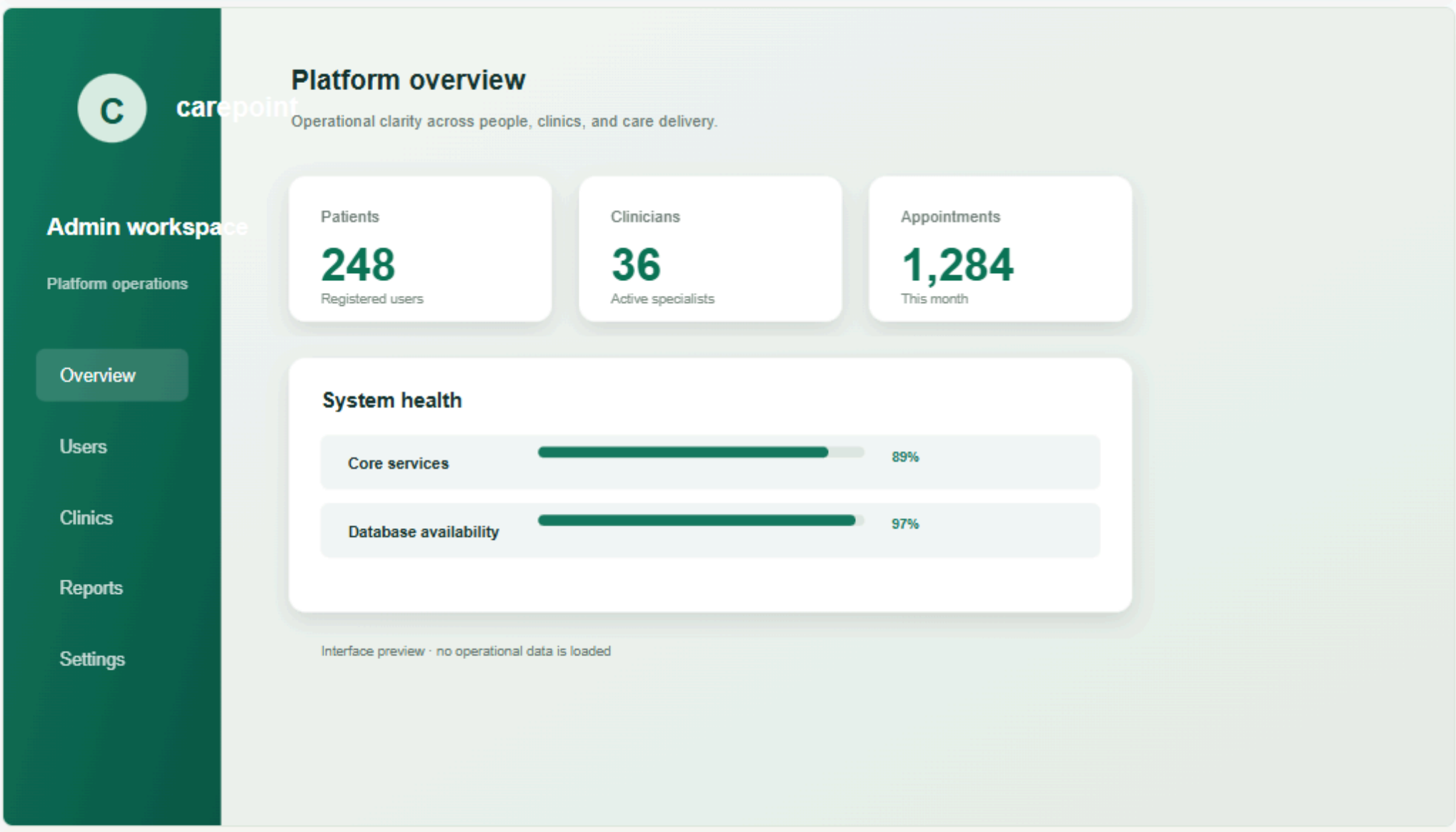
THE COMPLETE CARE JOURNEY

One connected experience.

The application supports discovery, booking, clinical documentation, follow-up and feedback across the care team.







Interface previews

These three dashboard visuals are intentionally labeled as mockups and contain no real patient or operational data.

SAFE BY DESIGN

Clear boundaries for every action.

The application applies role checks, input validation, secure passwords, CSRF protection and repository-based data access.

✓ **Role-based access**

Every protected route verifies the signed-in account's role.

✓ **Stored securely**

Passwords use salted PBKDF2 rather than plain-text storage.

✓ **Validated input**

Appointment, scheduling and profile data are checked before persistence.

✓ **CSRF protection**

State-changing requests require a valid session token.

✓ **Prepared SQL**

Repository queries use placeholders and domain-focused methods.

✓ **Unit tested**

Core password behavior is covered by JUnit tests.

READY FOR REVIEW

Build, test and present.

The Maven project can be compiled and tested, then served locally with Jetty for demonstration.

Build and verify

- `mvn clean test package`
- Verify Java compilation and JUnit results
- Run the packaged WAR with Jetty
- Confirm the database settings are available

Submission handoff

- Push the repository to a reviewer-accessible GitHub URL
- Include the application README and database schema
- Share this PDF and the screenshot collection
- Record the exact run instructions and demo credentials

Carepoint Healthcare Academic demonstration only. No real patient information is included in this presentation.

CAREPOINT / CURRENT FEATURE SET

Clinic operations, now end to end.

Recent updates complete the administrator workflows while preserving the same scheduling rules.

People & roles

Create patient, doctor, or administrator accounts.

Edit accounts; activate, deactivate, or delete when safe.

Accounts with linked clinic history are protected from deletion/role changes.

Schedule for a patient

Admins select an active patient and an active doctor.

Booking must fit published availability and pass conflict checks.

Reschedule uses the same availability and overlap validation.

Clinic settings

Set clinic name, public email, phone, and address.

Choose a new-appointment duration from 10 to 180 minutes.

Clinic identity/contact details are shown on public pages.

Operational analytics

View appointment volume by status and recent clinic visits.

Compare six months of appointment and account-registration trends.

Role-specific workspaces keep responsibilities clear.

CONTROL FLOW

JSP form -> role-checked servlet action -> JDBC repository -> MySQL

Built with clear boundaries.

The project uses standard Java web layers, persisted clinic data, and explicit request protections.

Web & domain

Java 17, Jakarta Servlets, JSP, Maven, Jetty 12.

Immutable Java records model users, visits, feedback, and metrics.

Servlets coordinate actions; repositories own JDBC persistence.

MySQL & transactions

Prepared statements and role-scoped queries protect data access.

Appointments link patients, doctors, and published availability.

Rescheduling and multi-setting updates use database transactions.

Security practices

Salted PBKDF2 password hashes; session ID rotation.

Role authorization, CSRF checks, validation, and security headers.

Prepared SQL avoids assembling user input into queries.

Verification

mvn clean test package: BUILD SUCCESS.

5 tests passed; database integration tests enabled against local MySQL.

Jetty smoke-tested: home, clinics, and role sign-in routes returned HTTP 200.

LOCAL DEMO

JDK 17+ | Maven 3.9+ | MySQL 8+ | mvn clean test package

Set DB_* values; use DB_INTEGRATION_TESTS=true to run MySQL integration tests.